Roadmap. Srednicki §§33–35: where spinors come from. Field representations $L_A{}^B$ and the algebra (pp. 1–4); $N_i, N_i^\dagger$ and the $(2n+1, 2n'+1)$ labels (pp. 4–6); Weyl fields ψ_a, $\psi_a^\dagger$, $S_R = -S_L^*$ (pp. 7–10); invariant symbols ε, σ^μ, $\bar\sigma^\mu$, index rules (pp. 11–16); pp. 16–17 answer p. 2's question. Next: Majorana and Dirac Lagrangians, γ^μ (LecNote17, §36). Supplements: FieldTransformations, LorentzGroup, SpinorRepresentations, DiracSpinors, FermionSigns.

We are very used to the vector-representation of the Lorentz group as space-time is a Lorentz 4-vector.
But other non-trivial representations exist and of particular importance, the representation naturally suitable for Fermions (Dirac's legendary contribution§, together with many others).

Although practically I can simply show you the matrices & the Feynman rules, which will enable some calculation I want, instead show you where the spinors come from, and its transformation under Lorentz Groups. You need to at least see it once in your life, as a physicist.

Unitary operator transforms a scalar field $\varphi(x)$

$$U(\Lambda)^{-1}\,\varphi(x)\,U(\Lambda) = \varphi(\Lambda^{-1}x)$$

$$U(\Lambda)^{-1}\,\partial^\mu\varphi(x)\,U(\Lambda) = \Lambda^\mu{}_\rho\,\bar\partial^\rho\,\varphi(\Lambda^{-1}x).$$

$$\frac{\partial}{\partial \bar x_\rho}, \quad \bar x \equiv \Lambda^{-1}x.$$

Or for generic vector field $A^\mu(x)$, $B^{\mu\nu}(x)$

$$U(\Lambda)^{-1}A^\mu(x)\,U(\Lambda) = \Lambda^\mu{}_\nu\,A^\nu(\Lambda^{-1}x)$$

$$U(\Lambda)^{-1}B^{\mu\nu}(x)\,U(\Lambda) = \Lambda^\mu{}_\rho\,\Lambda^\nu{}_\sigma\,B^{\rho\sigma}(\Lambda^{-1}x)$$

Generic tensor is decomposible to three component that do not mix under L.T.

$$B^{\mu\nu}(x) = A^{\mu\nu}(x) + S^{\mu\nu}(x) + \frac{1}{4} g^{\mu\nu} T(x)$$

$A^{\mu\nu} = -A^{\nu\mu}$ $\quad S^{\mu\nu} = S^{\nu\mu}$ $\quad$ Scalar. and ensures that $S^{\mu\nu}(x)$ is traceless.

$$U(\Lambda)^{-1} \frac{1}{4} g^{\mu\nu} T(x) U(\Lambda) = \frac{1}{4} g^{\rho\sigma} \Lambda^{\mu}{}_{\rho} \Lambda^{\nu}{}_{\sigma} T(\Lambda^{-1}x)$$

$$= \frac{1}{4} g^{\mu\nu} T(\Lambda^{-1}x).$$

Clearly the Lorentz transformations of $A^{\mu\nu}(x)$, $S^{\mu\nu}(x)$, $T(x)$ do not mix.

As they correspond to different symmetries or antisymmetries of the matrices.

Can it be further decomposed into unmixed transformations?

Generically, for a general Lorentz (contravariant) covariant state with an abstract index (subscript A).

(note that we did not choose μ as it is not necessarily the vector representation)

(Secretly, we all know the existence of fermions).

(We will briefly mention a different perspective later: the Dirac's heuristic method) (this should be covered in advanced QM or particle physics I).

We would have

$$U(\Lambda)^{\dagger}\varphi_A(x)\,U(\Lambda) = \underbrace{L_A{}^B(\Lambda)}_{\text{General transformation on }\Lambda.}\varphi_B(\Lambda^{-1}x)$$

We hope for a finite dimensional rep (matrix), to obey the group composition rules and hence forms a representation of the group.

$$L_A{}^B(\Lambda')\,L_B{}^C(\Lambda) = L_A{}^C(\Lambda'\Lambda)$$

For an infinitesimal transformation $\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \delta\omega^{\mu}{}_{\nu}$ we can write down that

$$U(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}\underbrace{M^{\mu\nu}}_{\uparrow}$$

(which we introduced at the beginning of this class)

Generators of the Lorentz group.

They obey the **Lie Algebra** of the Lorentz group:

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\mu\rho}M^{\nu\sigma} - (\mu\leftrightarrow\nu)) - (\rho\leftrightarrow\sigma)$$

We've already identified various physical components.

Angular momentum $J_i = \frac{1}{2}\varepsilon_{ijk}M^{jk}$

Boost operator $K_i \equiv M^{i0}$

with $[J_i, J_j] = i\varepsilon_{ijk} J_k$

$[J_i, K_j] = i\varepsilon_{ijk} K_k$

$[K_i, K_j] = -i\varepsilon_{ijk} J_k$

Now, we can write down the general index

$$L_A{}^B(1+\delta\omega) = \delta_A{}^B + \frac{i}{2}\omega_{\mu\nu}\underbrace{(S^{\mu\nu})_A{}^B}_{\text{another representation}}$$

To the ω order, the general transformation

$$[\varphi_A(x), M^{\mu\nu}] = \underbrace{\frac{1}{i}(x^\mu\partial^\nu - x^\nu\partial^\mu)}_{\mathcal{L}^{\mu\nu}\ (\Gamma\text{symbol}\rfloor)}\varphi_A(x) + (S^{\mu\nu})_A{}^B\varphi_B(x)$$

They need to satisfy the Lie Algebra, do we have other choices?

(Recall your knowledge of spin is not invariant under rotation of 2π; there is a minus sign. As the spin is represented by $SU(2)$; not $SO(3)$, double-cover...)

(One can work out ~~this~~ such reps for arbitrary dim, but that's a separate topic. Here we show one trick.)

Note one can define:

$$N_i \equiv \frac{1}{2}(J_i - iK_i)$$

$$N_i^\dagger \equiv \frac{1}{2}(J_i + iK_i)$$

We find that

$$[N_i, N_j] = i\epsilon_{ijk} N_k.$$
$$[N_i^\dagger, N_j^\dagger] = i\epsilon_{ijk} N_k^\dagger$$
$$[N_i, N_j^\dagger] = 0.$$

But $[\quad, \quad] = i\epsilon_{ijk}$ is for $SU(2)$.

So we found two "separate" $SU(2)$'s that are related by hermitian conjugation

$$N_i = (N_i^\dagger)^\dagger, \quad N_i^\dagger = (N_i)^\dagger$$

Note that $SU(2)$ are specified by integers and representations half integers. (Recall the rotation group for angular momentum, we have J^2 and J_z.)

Since J_i and K_i are all the ~~generators~~ associate with the Lorentz group (non-zero generators, $M^{\mu\nu}$ anti-symmetric Hermitian),

$$J_i = N_i + N_i^\dagger, \quad K_i = i(N_i - N_i^\dagger),$$

The two $SU(2)$'s also represent the Lorentz group. We have another representation!

~~And~~ And

a representation of the Lorentz group in 4D is fixed by two integers n and n' [1]

1 Should read: n, n' integers or half-integers (as above; p. 6's (2,1) has $n = \frac{1}{2}$); the labels $2n+1$, $2n'+1$ are the integers. For $n + n'$ half-integer a 2π rotation gives -1.

We label these representations as $(2n+1, 2n'+1)$ as it helps counting the # of components.

For a SU(2) with label n, the components are $2n+1$

So the total # of components for $(2n+1, 2n'+1)$ is $(2n+1)(2n'+1)$.

Different components within a representation can also be labled (grouped) by their angular momentum representations.

Since $J_i = N_i + N_i^+$

then the allowed j are $|n-n'|, |n-n'|+1, \cdots, n+n'$

The most simple & often encountered representations are

$(1,1)$ = ~~singlet~~ Lorentz Scalar / singlet

$(2,1)$ = left-handed spinor ← transfor under N_i

$(1,2)$ = right-handed spinor ← transfor under N_i^+

$(2,2)$ = vector.

Lorentz vector has 4 components and are irreducible (they all mix under rotation + boost).

$(4,1)$ and $(1,4)$ contain 4 components but $j = \frac{3}{2}$

$(2,2)$ contain 4 components, with $j=1$ and $j=0$

($j=1$: 3 components → 3 vector.) ($j=0$: 1 component → time.)

again it is all mixed as J_i, K_i transform both the 2 doublets.

Here we just ~~found some~~ found some interesting representations of the Lorentz group which contain "angular momentum"-like ~~eigen~~ states with $j=1/2$. Without any surprise, they are fermions. and they are special, that they are not labelled by vector index μ, but some abstract index A.

Let's study $(2,1)$ state, we call it left-handed Weyl field $\psi_a(x)$, back to our general expectation on their transformations.

$$U(\Lambda)^{-1}\psi_a(x)U(\Lambda) = \underbrace{L_a{}^b(\Lambda)}_{\text{a matrix in the } (2,1) \text{ rep.}}\psi_b(\Lambda^{-1}x)$$

following the group composition rule

$$L_a{}^b(\Lambda')L_b{}^c(\Lambda) = L_a{}^c(\Lambda'\Lambda)$$

For an infinitesimal transformation, we can write

$$L_a{}^b(1+\delta\omega) = \delta_a{}^b + \frac{i}{2}\delta\omega_{\mu\nu}(\underbrace{S_L^{\mu\nu}}_{\text{left-handed (so far we are just calling it, no physical meaning yet. Don't worry too much!)}})_a{}^b$$

Similar to our 1st HW, we can find that.

$$[S_L^{\mu\nu}, S_L^{\rho\sigma}] = i(g^{\mu\rho}S_L^{\nu\sigma} - (\mu\leftrightarrow\nu)) - (\rho\leftrightarrow\sigma)$$

~~Ast~~

Again $[\psi_a(x), M^{\mu\nu}] = \mathcal{L}^{\mu\nu}\psi_a(x) + (S_L^{\mu\nu})_a{}^b \psi_b(x)$

$\mathcal{L}^{\mu\nu} \to \frac{1}{i}(x^\mu\partial^\nu - x^\nu\partial^\mu)$

we do not focus on this term and it disappears when we evaluate the fields at the origin. ($x^\mu = 0$)

Recalling that ~~$J_k = \frac{1}{2}\varepsilon_{ijk} M^{ij}$~~

$J_i = \frac{1}{2}\varepsilon_{ijk} M^{jk}$, then

$M^{ij} = \varepsilon^{ijk} J_k$. and J is the angular momentum operator.

then.

$$\varepsilon^{ijk}[\psi_a(0), J_k] = (S_L^{ij})_a{}^b \psi_b(0)$$

We also know that a ~~spin~~ angular momentum $-1/2$ object transforms under J_k.

to be $\frac{1}{2}\varepsilon^{ijk}\sigma_k$ (from QM class).

Here σ is a Pauli Matrix

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

This leads to that

$$(S_L^{ij})_a{}^b = \frac{1}{2}\varepsilon^{ijk}\sigma_k.$$

↑ spin-1/2 ↑ more standard double-index notation. vs notation

For boost $K_i = M^{i0}$ and

$$K_i = i(N_i - N_i^\dagger)$$

but $(2,1)$ is a singlet on $N_i^\dagger$ so

$$(S_L^{i0})_a{}^b = \frac{1}{2} i\sigma_i \text{, so we get the full } (S_L^{\mu\nu})_a{}^b$$

(Note here σ_i are just matrices, not about Lorentz vector indices)

As $(N_i)^\dagger$ is $N_i^\dagger$,

we anticipate to get all the properties on $(1,2)$ state, right-handed spinor from Hermitian conjugation, to make clear that a state is the $(1,2)$, we use dotted index to note them.

$$[\psi_a(x)]^\dagger = \psi^\dagger_{\dot a}(x)$$

Under a Lorentz transformation, we have

$$U(\Lambda)^{-1}\psi^\dagger_{\dot a}(x)\,U(\Lambda) = R_{\dot a}{}^{\dot b}\,\psi^\dagger_{\dot a}(x)$$ (1)

1 Should read $R_{\dot a}{}^{\dot b}(\Lambda)\,\psi^\dagger_{\dot b}(\Lambda^{-1}x)$ (cf. p. 7).

Similarly

$$R_{\dot a}{}^{\dot b}(1+\delta\omega) = \delta_{\dot a}{}^{\dot b} + \frac{i}{2}\delta\omega_{\mu\nu}(S_R^{\mu\nu})_{\dot a}{}^{\dot b}$$

and

$\uparrow$ anti-symmetric matrices obey the Lie Algebra, similar to $M^{\mu\nu}$ and $S_L^{\mu\nu}$.

$$[\psi^\dagger_{\dot a}(0), M^{\mu\nu}] = (S_R^{\mu\nu})_{\dot a}{}^{\dot b}\,\psi^\dagger_{\dot b}(0)$$

Hermitian conjugate the above (Note $(S_R^{\mu\nu})_{\dot a}{}^{\dot b}$ is a number)

we get ($M^{\mu\nu}$ hermitian)

$$[M^{\mu\nu}, \psi_a(0)] = [(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}}]^* \psi_b(0)$$

but we already know

$$[\psi_a(0), M^{\mu\nu}] = (S_L^{\mu\nu})_a{}^b \psi_b(0)$$

so $[(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}}]^* = -(S_L^{\mu\nu})_a{}^b$ (1)

1 Two facts used: $M^{\mu\nu\dagger} = M^{\mu\nu}$ (U unitary) and $[A,B]^\dagger = [B^\dagger, A^\dagger]$, so $[\psi_{\dot{a}}^\dagger, M^{\mu\nu}]^\dagger = [M^{\mu\nu}, \psi_a] = -(S_L^{\mu\nu})_a{}^b \psi_b$. Finite: $R_{\dot{a}}{}^{\dot{b}} = (L_a{}^b)^*$. The daggers act on operators; as 2×2 matrices $K_i = \frac{i}{2}\sigma_i$ is anti-Hermitian and the $\dagger$ in $N_i^\dagger$ is a label (SpinorRepresentations).

component by component, we know $(S_R^{\mu\nu})$ and how right-handed Weyl field worded transform under Lorentz.

Let's digest briefly here:

(A). (2,1) and (1,2) are the lowest point-~~like~~ like non-singlet spin states we have. With such building blocks, we can represent any higher spins. It is just often much more convient to use vector-reps, instead of spinor-reps for states with integer spins. (e.g. vector, tensor, etc.) But for fermions, ~~with~~ including spin-$3/2$, $5/2$, ... we have no choices but to use spinor-rep.

Spinor-rep can also represent vectors, and in fact, shed some insights for us as well.

(13). Typical operations, rotation J and boost K, now correspond to a combination of N_i and $N_j^\dagger$, which means, if I rotate J by $\delta\theta$, both the (1,2) and ~~(2,1)~~ (2,1) states will rotate in a correlated way.

Now, back to discover more properties for such reps. Let's explore a field with two (2,1) indices, $C_{ab}(x)$

$$U(\Lambda)^\dagger C_{ab}(x)\, U(\Lambda) = L_a{}^c(\Lambda)\, L_b{}^d(\Lambda)\, C_{cd}(\Lambda^{-1}x).$$

This 4 (2×2) component state can be further decomposed. We already know, they can make spin 0 (antisymmetric) or spin-1 (symmetric). $2 \otimes 2 = 1_A \oplus 3_S$.

For Lorentz group, we have

$$(2,1) \otimes (2,1) = (1,1)_A \oplus (3,1)_S.$$

we should be able to write

$$C_{ab}(x) = \varepsilon_{ab} D(x) + G_{ab}(x)$$

$D(x)$: spinless, Lorentz Scalar.

$\varepsilon_{ab} = -\varepsilon_{ba}$, $\varepsilon_{21} = -\varepsilon_{12} = 1$

$G_{ab}(x)$: $G_{ab}(x) = +G_{ba}(x)$ as it is antisymmetric. (1)

1 Should read "symmetric": G_{ab} is the $(3,1)_S$ part; the antisymmetric part is $\varepsilon_{ab}D$.

This implies that ε_{ab} is an invariant symbol of the Lorentz group. (2)

2 Why: for any 2×2 matrix, $L_a{}^c L_b{}^d \varepsilon_{cd} = (\det L)\,\varepsilon_{ab}$, and $\det L = \exp[\frac{i}{2}\omega_{\mu\nu}\mathrm{tr}\, S_L^{\mu\nu}] = 1$ since S_L (pp. 8–9) is traceless: $L \in SL(2,\mathbb{C})$. Likewise $R = L^*$, so $\varepsilon_{\dot a\dot b}$ (p. 13) is invariant. The antisymmetric part of C_{ab} has one component, hence $\varepsilon_{ab}D$.

$$L_a{}^c(\Lambda) L_b{}^d(\Lambda) \varepsilon_{cd} = \varepsilon_{ab}$$

We already have another invariant symbols, and we will find more.

$$\Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma g_{\rho\sigma} = g_{\mu\nu}$$

We use $g_{\mu\nu}$ and $g^{\mu\nu}$ to raise and lower indices, and we can do so as well for ε_{ab} and ε^{ab} for the left-handed indices.

$$\varepsilon^{12} = \varepsilon_{21} = +1, \quad \varepsilon^{21} = \varepsilon_{12} = -1 \qquad \text{(definitions)}$$

Hence we will have

$$\varepsilon_{ab}\varepsilon^{bc} = \delta_a{}^c, \quad \varepsilon^{ab}\varepsilon_{bc} = \delta^a{}_c$$

Then we have: $\psi^a(x) \equiv \varepsilon^{ab}\psi_b(x)$, it's consistently

$$\psi_a(x) = \varepsilon_{ab}\psi^b(x) = \varepsilon_{ab}\varepsilon^{bc}\psi_c(x) = \delta_a{}^c\psi_c(x) = \psi_a(x)$$

But ε^{ab} is anti-symmtric, we need to be careful with signs

$$\psi^a(x) = \underbrace{\varepsilon^{ab}}\psi_b(x) = -\varepsilon^{ba}\psi_b(x) = \underbrace{-\psi_b(x)\varepsilon^{ba}} = \psi_b(x)\varepsilon^{ab}$$

↑ view it as an element in matrices and then summed over.

$$\psi^a\chi_a = \varepsilon^{ab}\psi_b\chi_a = -\varepsilon^{ba}\psi_b\chi_a = -\psi_b\chi^b$$

So even with contracted indices, "$-$" sign appears if one is not careful enough.

Again, we show these basics, which later on in more practical cases, there are simplifications.

1 Why (2,2) is the vector: for real A_μ, $A_{a\dot{a}} = A_\mu \sigma^\mu_{a\dot{a}}$ is a Hermitian matrix with $\det A = -A^\mu A_\mu$. It transforms as $LAR^T = LAL^\dagger$ ($R = L^*$, p. 10), which keeps Hermiticity and, as $\det L = 1$, the determinant: a Lorentz transformation of A^μ. That is p. 14's invariance of $\sigma^\mu_{a\dot{a}}$ (SpinorRepresentations).

Similarly $(1,2)\otimes(1,2) = (1,1)_A \oplus (1,3)_S$.
and we find the invariant symbol
$\varepsilon_{\dot{a}\dot{b}} = -\varepsilon^{\dot{a}}{}_{\dot{b}\dot{a}}$, with a same convention to take
$\varepsilon^{12} = \varepsilon_{21} = 1$, $\varepsilon^{21} = \varepsilon_{12} = -1$

Now turn to $(2,2)$, which is a vector. How? (1)
Need a dictionary to translate / project
double spinor index (one left one right) to vector index μ.

$$A_{a\dot{a}}(x) = \underset{\text{should be}}{\sigma^\mu_{a\dot{a}}} A_\mu(x), \text{ here } \sigma^\mu_{a\dot{a}} = (I, \vec{\sigma})$$

as it is a general translation between $a\dot{a}$ $\longleftrightarrow$ to μ
One can construct it through another singlet, which is proven in the beginning part of [See 35.] (I only highlight a few parts here).
If such an invariant symbol exists, there can be various invariant products.

$$\sigma^\mu_{a\dot{a}}\sigma_{\mu b\dot{b}} = -2\varepsilon_{ab}\varepsilon_{\dot{a}\dot{b}} \quad \longleftarrow \text{exactly what you anticipate to form a singlet}$$
$$\varepsilon^{ab}\varepsilon^{\dot{a}\dot{b}}\sigma^\mu_{a\dot{a}}\sigma^\nu_{b\dot{b}} = -2g^{\mu\nu}$$
$(2,1)\otimes(1,2)\otimes(2,2) = (1,1)\oplus\ldots$
or
$(2,2)\otimes(2,2) = (1,1)\oplus\ldots$

one can translate between different invariant symbols.

And one can find and verify that, the generators
$(S^{\mu\nu}_L)_{ac}$, $(S_R{}^{\mu\nu})_{\dot{a}\dot{c}}$ and also be expressed as

$$(S_L^{\mu\nu})_{ae} = \frac{i}{4}\varepsilon^{\dot{a}\dot{c}}(\sigma^\mu_{a\dot{a}}\sigma^\nu_{c\dot{c}} - \sigma^\nu_{a\dot{a}}\sigma^\mu_{c\dot{c}})$$

$$(S_R^{\mu\nu})_{\dot{a}\dot{c}} = \frac{i}{4}\varepsilon^{ae}(\sigma^\mu_{a\dot{a}}\sigma^\nu_{c\dot{c}} - \sigma^\nu_{a\dot{a}}\sigma^\mu_{c\dot{c}})$$

↑

coming from using $\sigma^\mu_{a\dot{a}}$ being invariant symbol

$$\sigma^\mu_{a\dot{a}} = \Lambda^\mu{}_\nu\, L(\Lambda)_a{}^b\, R(\Lambda)_{\dot{a}}{}^{\dot{b}}\, \sigma^\nu_{b\dot{b}}$$

↓ can be written as

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \frac{i}{2}\delta\omega_{\rho\sigma}\left(\frac{1}{i}\right)\left(g^{\rho\mu}\delta^\sigma{}_\nu - g^{\sigma\mu}\delta^\rho{}_\nu\right)$$

~~Conventionally~~ Conventionally, one can define another invariant symbol

$$\bar{\sigma}^{\mu\dot{a}a} \equiv \varepsilon^{ab}\varepsilon^{\dot{a}\dot{b}}\sigma^\mu_{b\dot{b}}$$

$$\bar{\sigma}^{\mu\dot{a}a} = (I, -\vec{\sigma})$$

Now, more concisely.

$$(S_L^{\mu\nu})_a{}^b = +\frac{i}{4}(\sigma^\mu{}^{a\dot{c}}\bar{\sigma}^{\nu\dot{c}b} - \sigma^\nu{}^{a\dot{c}}\bar{\sigma}^{\mu\dot{c}b})_a{}^b$$

$\dot{c}b$ contracted (matrix multiplication)

$a\dot{c}$ contracted.

$$(S_R^{\mu\nu})^{\dot{a}}{}_{\dot{b}} = -\frac{i}{4}(\bar{\sigma}^{\mu\dot{a}c}\sigma^\nu - \bar{\sigma}^{\nu\dot{a}c}\sigma^\mu)^{\dot{a}}{}_{\dot{b}}$$

Taking this contraction convention, dotted are contracted as $\dot{c}^{\dot{c}}$ and undotted as ${}^c{}_c$, we get useful simplification that.

1 Links: $\mu\nu = i0$ gives $\frac{i}{2}\sigma_i$, ij gives $\frac{1}{2}\varepsilon^{ijk}\sigma_k$ (pp. 8–9). The top lines lower the second index, $(S_L)_{ac} = \varepsilon_{cb}(S_L)_a{}^b$; $(S_R)^{\dot{a}}{}_{\dot{b}}$ is p. 10's S_R with both indices moved by ε. The symmetric part, $\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2g^{\mu\nu}$ (same for $\bar{\sigma}\sigma$), gives $\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}$ in LecNote17 p. 2.

$\chi\psi = \chi^a \psi_a$ and $\chi^\dagger \psi^\dagger = \chi^\dagger_{\dot a} \psi^{\dagger \dot a}$

↑ ↑
left-handed fields

We also anticipate fermion fields are odd under exchange.

$$\psi_a(x)\chi_b(y) = -\chi_b(y)\psi_a(x)$$

Now we have

$$\chi\psi = \chi^a\psi_a = -\psi_a\chi^a = \psi^a\chi_a = \psi\chi$$

↑ exchange get a minus

↑ a^a to $^a a$ gets a minus sign as shown earlier.

Same for $(\chi\psi)^\dagger = \psi^\dagger\chi^\dagger = \chi^\dagger\psi^\dagger$ as per our conventions

Further, one can adopt a convention that the a right handed field always written as the Hermitian conjugate of a left-handed field. (and left-handed fields are written without a dagger)

for instance.

$$\psi^\dagger_{\dot a}\bar\sigma^{\mu \dot a a}\chi_a \xrightarrow{\text{now can be written as}} \psi^\dagger \bar\sigma^\mu \chi$$

$\dot a$ and a are independent indices

All the spinor index contracted and now with only a bare vector index μ, it will transform like a vector.

Hence

$$U(\Lambda)^{-1}[\psi^\dagger \bar{\sigma}^\mu \chi]U(\Lambda) = \Lambda^\mu{}_\nu [\psi^\dagger \bar{\sigma}^\nu \chi]$$

$\psi^\dagger(x)$ $\chi(x)$ $\psi^\dagger(\Lambda^{-1}x)$ $\chi(\Lambda^{-1}x)$

And

$$[\psi^\dagger \bar{\sigma}^\mu \chi]^\dagger = [\psi^\dagger_{\dot{a}} \bar{\sigma}^{\mu \dot{a}a} \chi_a]^\dagger$$

$$= \chi^\dagger_{\dot{a}} (\bar{\sigma}^{\mu \dot{a}a})^\dagger \psi_a$$

$$= \chi^\dagger_{\dot{a}} \bar{\sigma}^{\mu \dot{a}a} \psi_a \quad \text{as } \bar{\sigma}^\mu = (I, -\vec{\sigma}) \text{ is hermitian.}$$

$$= \chi^\dagger \bar{\sigma}^\mu \psi$$

We are going to build ~~for~~ spinor Lagrangian with these properties and further notational simplifications. (by adopting conventions)

Before we get to QFT for ~~spin~~ fermions, we also solved some ~~some~~ small ~~problem~~ questions: can

$B^{\mu\nu}$ be further decomposed into different pieces that transform separately under Lorentz group?

$$B^{\mu\nu} = S^{\mu\nu} + A^{\mu\nu} + \frac{1}{4} g^{\mu\nu} T$$

$A^{\mu\nu}$ can be further decomposed.

B/μν under spinor representation is

$$\underbrace{(2,2)}_{F} \otimes \underbrace{(2,2)}_{F} = \underbrace{(1_A, 1_A)_S}_{\frac{1}{2}g^{\mu\nu}} \oplus \underbrace{(1_A, 3_S)_A \oplus (3_S, 1_A)_A}_{A^{\mu\nu}} \oplus \underbrace{(3_S, 3_S)_S}_{S^{\mu\nu}}$$

$\downarrow a\dot{a} \quad \downarrow \dot{a}a$

The projection operator (covariant)

$$(S_L^{\mu\nu})_a{}^b = -\frac{i}{2}\varepsilon^{\mu\nu\rho\sigma}(S_{L\rho\sigma})_a{}^b$$

$$(S_R^{\mu\nu})_{\dot{a}}{}^{\dot{b}} = -\frac{i}{2}\varepsilon^{\mu\nu\rho\sigma}(S_{R\rho\sigma})_{\dot{a}}{}^{\dot{b}}$$ (1)

$$A^{\mu\nu}(x) = \underbrace{G^{\mu\nu}(x)}_{\text{self-dual antisymmetric tensor}} + \underbrace{G^{\dagger\mu\nu}}_{\text{anti-self-dual antisymmetric tensor}}$$ (2)

$$G^{\mu\nu}(x) \equiv (S_L^{\mu\nu})^{ab} G_{ab}(x)$$

$$= -\frac{i}{2}\varepsilon^{\mu\nu\rho\sigma} G^{\dagger}_{\rho\sigma}(x)$$ (3)

$$G^{\dagger\mu\nu}(x) = -(S_R^{\mu\nu})^{\dot{a}\dot{b}} G_{\dot{a}\dot{b}}(x)$$

$$= +\frac{i}{2}\varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}(x)$$

1 Should read $+\frac{i}{2}$: $S_R = -S_L^*$ (p. 10) and ε is real, so the sign flips (as in $G^{\dagger\mu\nu}$ below). Both lines take Srednicki's $\varepsilon^{0123} = +1$ ($\varepsilon_{0123} = -1$, §34), as LecNote17's γ_5 formulas do.

2 What the split is: $(*A)^{\mu\nu} \equiv \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}A_{\rho\sigma}$ commutes with Lorentz transformations and $** = -1$, so $G = \frac{1}{2}(A - i*A)$ ($*G = iG$, (3,1)) and $G^\dagger = \frac{1}{2}(A + i*A)$ (1,3) never mix. $S_L^{\mu\nu}$ is self-dual: $S_L^{\mu\nu}A_{\mu\nu}$ sees only G. For $F^{\mu\nu}$: $G^{0i} = \frac{1}{2}(E - iB)^i$. The split needs i: over the reals $A^{\mu\nu}$ is irreducible (p. 1).

3 Should read $-\frac{i}{2}\varepsilon^{\mu\nu\rho\sigma}G_{\rho\sigma}$, no dagger: it follows from the S_L line above; self-duality relates G to itself.